Université Paris Dauphine - PSL

Master 2 Statistical and Financial Engineering

# Data Visualization Project with R

## How to explain electricity prices?

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# Contents

# 1 Introduction

## 1.1 Context and motivation

Electricity prices result from a complex equilibrium between numerous economic, climatic, energy-related, and geopolitical factors. In the short term, local weather conditions can simultaneously influence electricity demand (temperature, rainfall) and production supply (wind, solar, hydro). In the longer term, structural dynamics such as the energy transition or climate change profoundly reshape national generation mixes. In addition, electricity prices depend heavily on the cost of the commodities used for generation (gas, coal, lignite, carbon), as well as on cross-border trade between European countries through integrated, dynamic markets. These mechanisms are particularly pronounced in Europe, where national grids notably those of France and Germany are tightly interconnected. All of these interactions make electricity price modeling especially delicate and call for a multifactorial approach.

## 1.2 Project objective and research question

The goal of the challenge is to explain the daily variation of short-maturity (24-hour) electricity futures prices for two European countries: France and Germany. This is not a forecasting problem in the strict sense, but a statistical-explanation problem, aiming to relate observed price fluctuations to a set of concurrent explanatory variables.
More precisely, the task is to build a model able to estimate the daily variation of electricity futures prices from weather data, energy data (commodities, generation by source) and trade data (consumption, imports, exports, and bilateral exchanges). Model performance is evaluated using the Spearman correlation between predicted values and the actually observed variations, computed on the **pooled** France + Germany test set a point that, as section 4.3 will show, directly shapes the modeling choices made in this report.

## 1.3 Approach and outline of the report

The report follows a five-step approach. Section 2 presents the data and its cleaning. Section 3 explores the data to derive a small number of economic hypotheses about price formation hypotheses that then directly motivate the choice of features and model. Section 4 formalizes the modeling approach: it starts from a per-country reference model, identifies a structural limitation with respect to the challenge's official metric, and explains how that limitation was corrected to arrive at the final retained model. Section 5 reports that model's performance in cross-validation, on a frozen out-of-sample holdout, and on the challenge platform together with the resulting variable importance. Section 6 documents two new variables tested after the initial submission, only one of which proved useful. The report closes with a discussion of limitations and possible extensions.

# 2 Data

## 2.1 Source and description

The data is provided by QRT, available at this link, and covers two geographical areas: France (FR) and Germany (DE). The training set (`X_train`) contains 1494 observations, and the test set (`X_test`) contains 654. Each row represents one day for a given country. The explanatory variables (35 columns) fall into four categories:

1. **Weather:** Temperature (`TEMP`), Rain (`RAIN`), Wind (`WIND`).
2. **Energy generation:** Changes in production costs by source (Gas, Coal, Nuclear, Solar, Wind, etc.).
3. **Commodity markets:** Daily changes in global prices (European Gas `GAS_RET`, Coal `COAL_RET`, Carbon `CARBON_RET`).
4. **Usage and exchanges:** Total consumption, residual load (`RESIDUAL_LOAD`), and net imports/exports (`NET_IMPORT`, `DE_FR_EXCHANGE`).

The target variable (`TARGET`) is the daily variation of the electricity futures price.

## 2.2 Cleaning and preprocessing

A systematic completeness analysis was carried out to identify variables with missing values. As these were infrequent and mostly linked to occasional unavailability of exogenous variables, a treatment specific to their economic nature was applied. Weather and cross-border exchange variables were imputed by linear interpolation, to preserve the physical continuity of the underlying phenomena. Net flows, being more volatile, received a more conservative treatment combining short-gap interpolation with moving-average smoothing for the remaining gaps. Finally, missing values at the start or end of a series were corrected by forward/backward propagation of the last available observation, ensuring complete and temporally consistent series.

These treatments yield a fully usable dataset, free of artificial breaks, providing a solid basis for exploratory analysis and modeling.

# 3 Exploratory analysis: five hypotheses on price formation

Rather than a plain inventory of charts, this section is organized as a sequence of questions put to the data, each leading to a concrete economic hypothesis. These hypotheses (H1 to H5) are picked up explicitly again in section 3.6, where they are translated into concrete choices of features and model.

## 3.1 H1 – Both markets share the same shape of risk, at different intensities

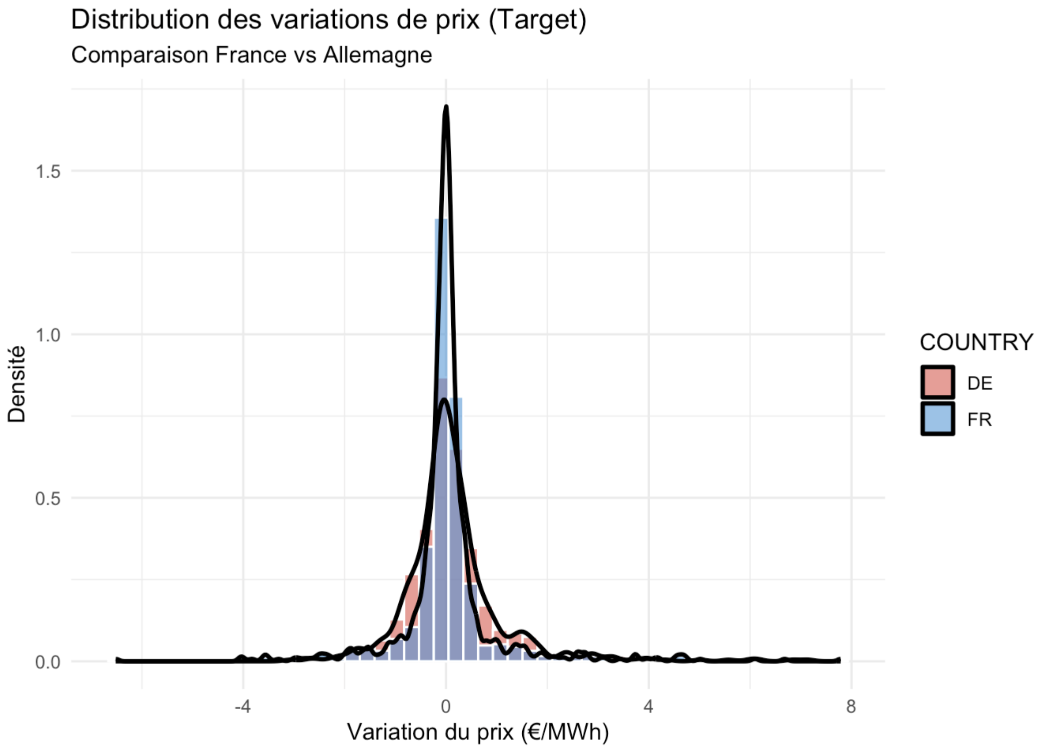



Figure 1: Distribution of price variations (`TARGET`) for France and Germany

The distribution of the target variable immediately reveals a feature common to both markets: price variations centered around zero but with fat tails, a sign that extreme events are not rare in these markets. This is more pronounced on the German side, where the dispersion is noticeably larger. **Hypothesis H1:** the German market is structurally more volatile than the French market, which should translate into a signal that is easier to rank (hence a better Spearman score) but also into a greater need for methods that are robust to extreme values. The following sections focus on understanding the mechanisms behind this asymmetry rather than simply restating it.

### 3.2 H2 – Physical availability and marginal costs form two independent forces

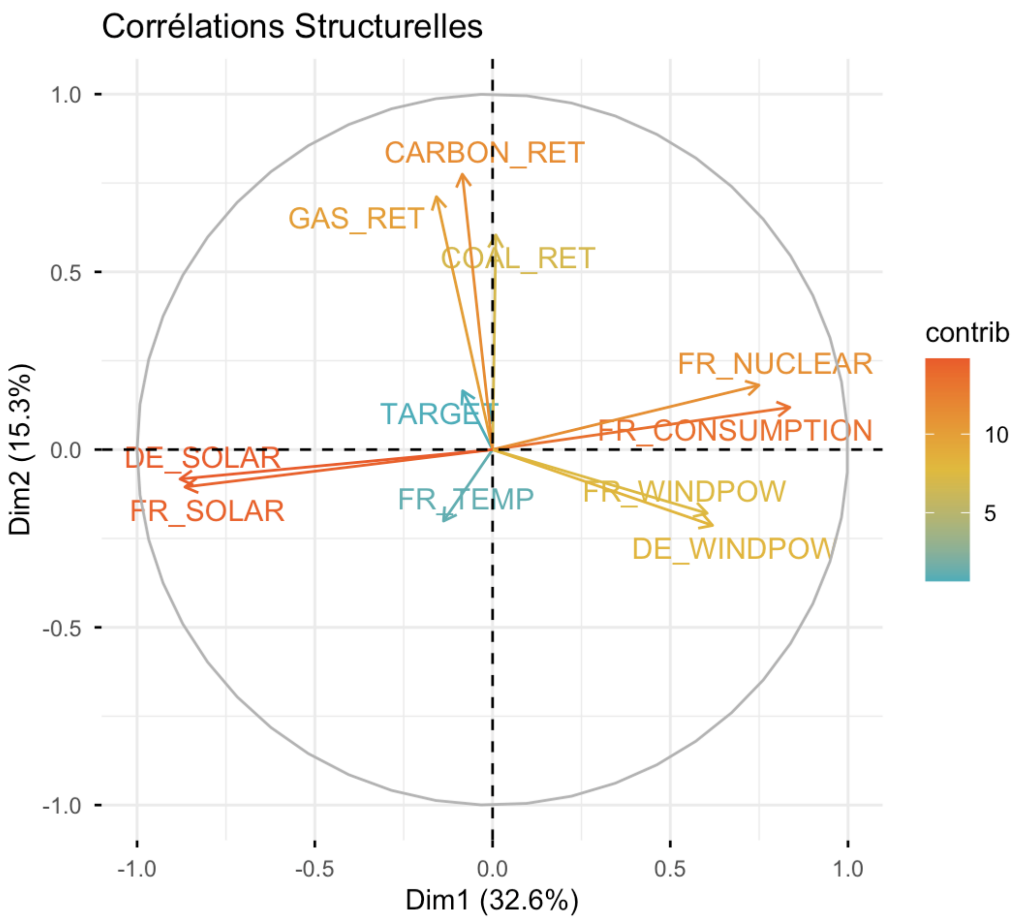



Figure 2: Principal Component Analysis of the physical and economic variables

A Principal Component Analysis (PCA) was carried out to summarize the dependence structure among the main physical and economic variables of the power system, while reducing the dimensionality of the problem. The variables included consumption, renewable generation (wind and solar), dispatchable generation (nuclear), weather conditions, and fossil fuel and carbon prices, so as to capture both the physical constraints of the grid and market signals. What clearly emerges is a structural opposition: renewable energy on one side, load and dispatchable capacity on the other. The second component mainly captures fuel price signals (gas, coal, carbon). **Hypothesis H2:** electricity prices are shaped by two relatively independent forces physical availability (renewable supply vs. load) and marginal costs (fuels) which argues for feature engineering that keeps these two families of variables separate rather than merging them into a single composite indicator.

## 3.3 H3 – The market alternates between discrete regimes rather than following a continuous dynamic

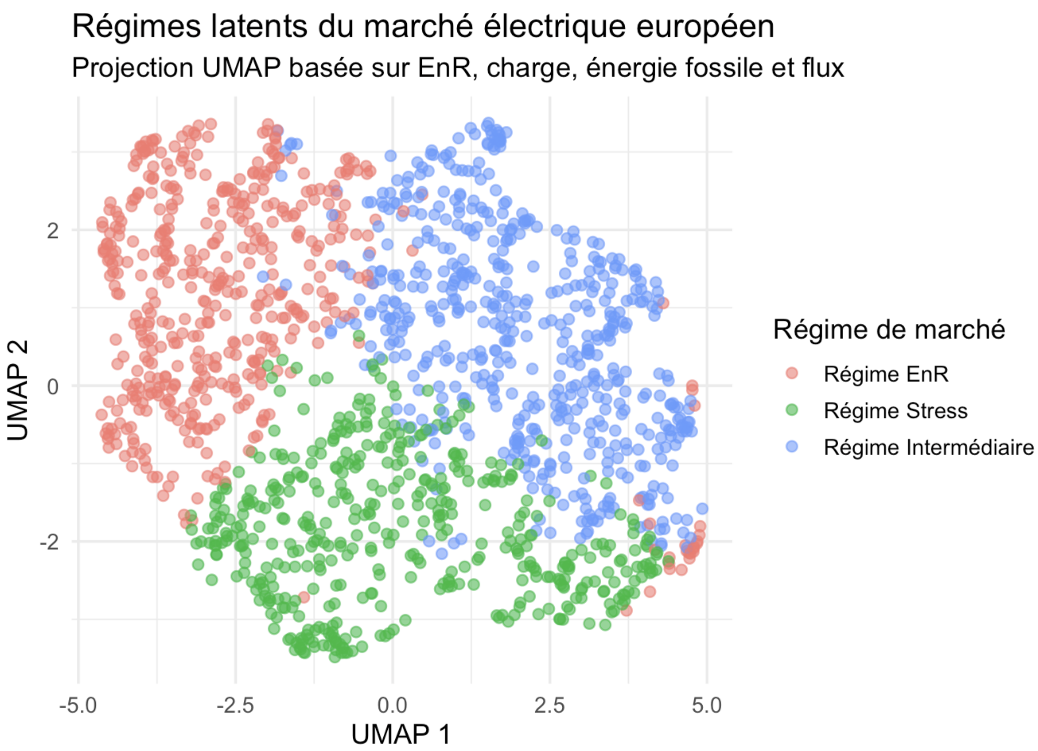



Figure 3: Latent regimes of the European electricity market

Clustering analysis reveals that the electricity market does not follow a linear trajectory, but alternates between three latent regimes with distinct structural dynamics:

- **Renewable regime:** massive wind and solar generation that saturates supply. Prices are strongly compressed, even negative, as zero-marginal-cost energy crowds out conventional thermal plants.
- **Intermediate regime:** standard market conditions, with moderate demand covered by a diversified energy mix. Prices are stable, reflecting a healthy balance between load and available generation capacity.
- **Stress regime:** high residual load (strong consumption combined with low renewable generation). The market must then call on costly generation units (gas, coal), driving prices up along with increased volatility, in line with *Merit Order* theory.

**Hypothesis H3:** the market does not follow a continuous dynamic but switches between qualitatively distinct regimes; a uniform model that is indifferent to this state change is misspecified it requires either non-linear features or an explicit regime indicator.

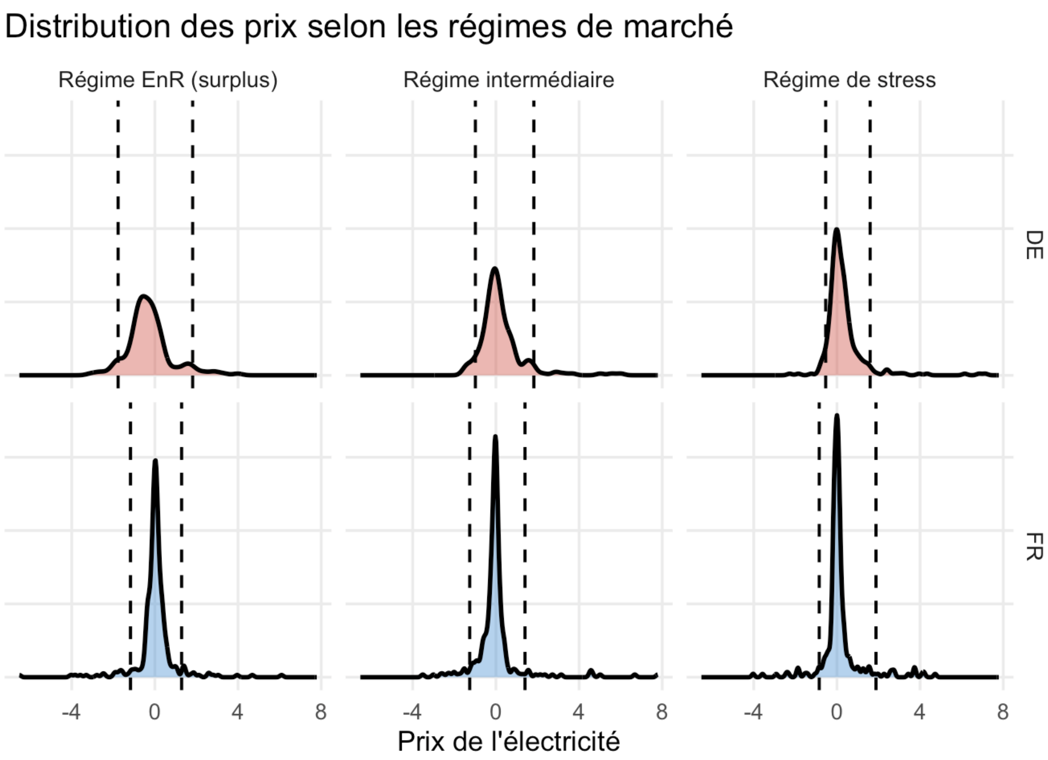
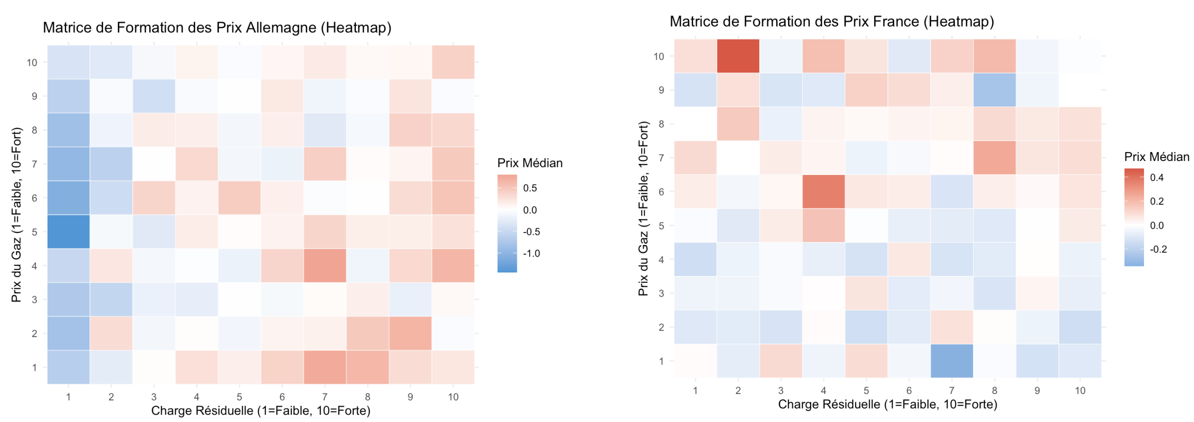



Figure 4: Distribution of prices conditional on market regime

Conditioning prices on the identified regimes shows that each regime has its own distribution "signature": the renewable regime produces tight, sometimes negative, prices; the stress regime generates a much wider spread with large spikes. This is not simply a matter of a different mean the very shape of the distribution changes from one regime to another a direct confirmation of H3, with implications for risk management as much as for model choice.

## 3.4 H4 – The residual load / gas price interaction is non-linear




(a) Germany

(b) France

Figure 5: Price-formation matrices by residual load and gas price

These matrices show how the price responds to the combination of residual load and gas price. The relationship is clearly non-linear: prices really spike only when both variables are simultaneously high, which is consistent with *Merit Order* logic. This interaction is more "aggressive" in Germany the high-price zones are more extensive and more pronounced whereas in France the surface is smoother, more dampened. **Hypothesis H4:** the relevant signal is not residual load or the gas price taken separately, but their interaction; an explicit interaction term (rather than relying on a linear model to discover it on its own) should improve the capture of this effect, more strongly in Germany than in France.

## 3.5 H5 – Renewables shift the mean price but also amplify its uncertainty

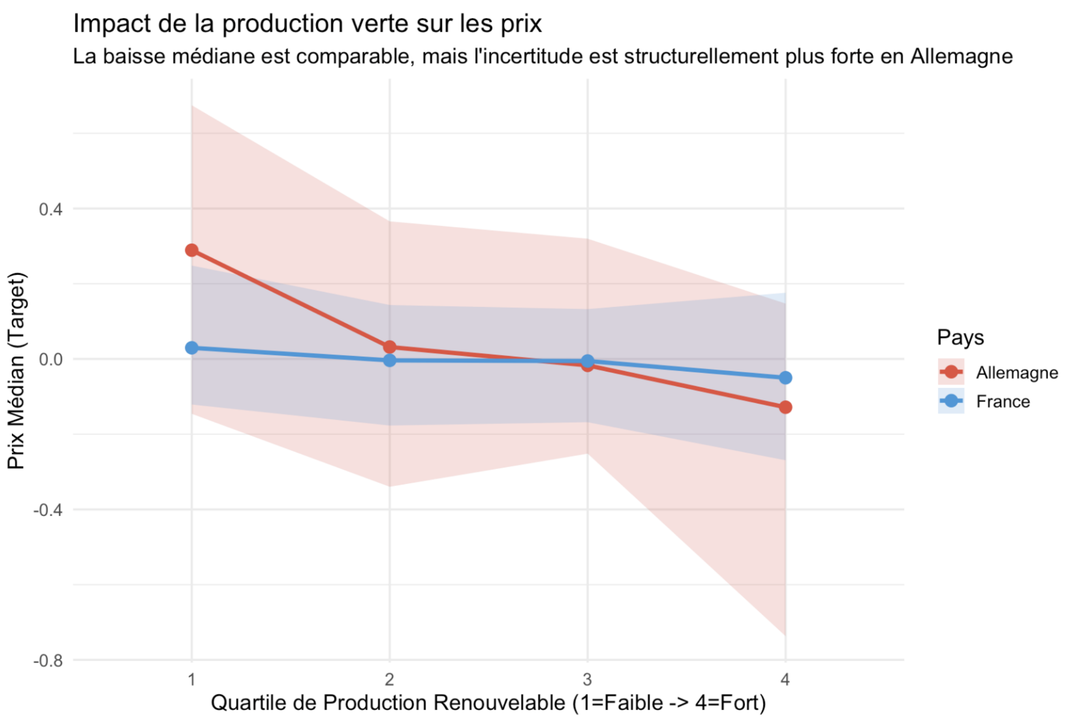



Figure 6: Impact of renewable generation on prices (by quartile)

This last analysis quantifies the downward effect of renewables on prices, as expected from *Merit Order* theory. The median effect is present in both countries. But the dispersion around that median increases sharply with renewable generation, especially in Germany. **Hypothesis H5:** the effect of renewables is not limited to shifting the central tendency of prices it also widens their conditional distribution an effect that a model capturing only the mean could easily miss.

### 3.6 Summary: from hypotheses to modeling choices

Table 1 summarizes how each of the five hypotheses above is concretely translated into the rest of the report, in section 4.

| Hyp. | Observation (section 3) | Translation into modeling (section 4) |
|---|---|---|
| H1 | Fat tails, DE more volatile than FR (3.1) | Target replaced by its rank (robustness to extreme values); higher scores expected for DE |
| H2 | Two independent forces: physical supply / marginal costs (PCA) | The two families of variables kept separate rather than merged into a composite index |
| H3 | Three discrete regimes, not a continuous dynamic (clustering) | Regime indicator (`REGIME_EnR`, `REGIME_Stress`) as an explanatory variable |
| H4 | Non-linear interaction residual load $\times$ gas (matrices) | Explicit interaction features (thermal stress, regime $\times$ fuel-switch spread) |
| H5 | Renewables widen dispersion too, not just the mean | Renewable-generation ratios and supply-saturation (*high-surplus*) indicators |

Table 1: From exploratory-analysis observations to feature-engineering and modeling choices

# 4 Methodology

## 4.1 Feature engineering

In line with the summary in table 1, a set of derived variables was built from the raw data. Temporal variables were first introduced as lags of the target (1, 2, and 24 time steps) along with rolling statistics (24-period mean and volatility), to capture short-term inertia and the persistence effects typical of electricity markets (memory and volatility-clustering phenomena, echoing H1).

The set of explanatory variables was also adapted separately for France and Germany (H2), to account for the structural differences in their generation mixes: certain German variables (wind, solar, residual load, flows) were explicitly included in the French model to capture market-coupling effects, while French variables that were less informative for Germany were dropped. Finally, non-linear transformations and interaction terms were introduced to model the underlying economic mechanisms (H3, H4):

quadratic temperature effects, thermal-stress indicators combining residual load and fuel costs, renewable-generation ratios relative to consumption, and supply-saturation indicators (H5). These variables reproduce the *Merit Order* mechanism, in which price is set by the marginal technology called upon, and improve the model's ability to discriminate extreme market regimes.

## 4.2 Feature selection

Variable selection was carried out in two complementary steps. First, the exploratory analysis of section 3 identified the physically and economically relevant variables for each country, motivating the construction of asymmetric variable sets between France and Germany (H2 above). Second, an automated cleaning procedure was applied before model training: near-zero-variance variables were removed, and multicollinearity was addressed by dropping variables that were highly correlated with one another (Spearman correlation above 0.8). This step reduces redundancy between very similar indicators, stabilizes model training, and improves its ability to generalize. The final selection thus retains a set of variables that is informative, interpretable, and non-redundant.

## 4.3 Modeling trajectory: from a per-country model to the retained model

The challenge's official metric computes a single Spearman correlation on the **pooled** France + Germany test set (section 1.2). This constraint, easy to lose sight of when models are evaluated separately by country, turned out to be decisive: it is the reason the modeling approach below unfolds in several steps rather than as a single model.

### 4.3.1 Reference model: per-country XGBoost

The reference model relies on the *Gradient Boosting* algorithm XGBoost, trained separately for France and Germany, which approximates the price function as an additive sum of weak learners (decision trees). Formally, the prediction is written:

$$\hat{y}_i = \sum_{m=1}^{M} f_m(x_i), \quad f_m \in \mathcal{F},$$

where each $f_m$ is a decision tree belonging to the function space $\mathcal{F}$. At each iteration, XGBoost adds a new tree by minimizing an objective function made up of a loss term and a regularization term:

$$\mathcal{L} = \sum_{i=1}^{n} \ell(y_i, \hat{y}_i) + \sum_{m=1}^{M} \Omega(f_m), \quad \text{with} \quad \Omega(f) = \gamma T + \frac{1}{2}\lambda \|w\|^2,$$

where $T$ is the number of leaves in the tree and $w$ the associated weights. In line with H1 (fat tails, extreme values), the target variable is replaced by its rank so as to favor the relative structure of prices rather than their levels:

$$\ell(y_i, \hat{y}_i) = (\mathrm{rank}(y_i) - \hat{y}_i)^2, \qquad \rho_S = \mathrm{corr}\left(\mathrm{rank}(y), \mathrm{rank}(\hat{y})\right).$$

This strategy aligns model training with the goal of ranking market situations, which is particularly relevant for prices characterized by skewed distributions, fat tails, and a strong dependence on the operating regime of the power system (H1, H3).

#### 4.3.2 Limitation identified: the pooled-score problem

Nothing guarantees a priori that the outputs of two XGBoost models trained independently, one per country, live on a comparable scale. Two models can each rank days very well *within* their own country (hence the separate scores of 0.16 and 0.36) without their outputs, once mixed, producing a coherent pooled ranking which is exactly what the challenge metric measures. The reference model of section 4.3.1, evaluated only country by country, does not address this: its per-country scores (reported in section 5.1) therefore do not directly tell us about the pooled performance actually assessed by the platform.

#### 4.3.3 Four-model ensemble and recalibration onto the physical scale

To fix this scale problem, the per-country model was first extended into an ensemble of four approaches (linear regression, regularized elastic-net regression, XGBoost, LightGBM), weighted by their cross-validated performance. Each model is trained not on `TARGET` itself but on its rank within the country and fold considered (the "rank trick"), a transformation that is robust to extreme values (H1). The predicted rank is then recalibrated onto the country's physical scale via its empirical quantile function, which makes the two countries' outputs comparable once pooled.

#### 4.3.4 Retained model: rank shared across countries

Rather than fixing the scale problem after the fact, a more direct approach computes the rank *once, over both countries combined*, at each training fold. Each per-country model (an elastic-net) then learns directly to predict a position on a shared scale, with no recalibration needed. This approach also fixes a temporal-alignment flaw in the original folds (the France and Germany folds did not cover exactly the same period), by building folds from pooled `DAY_ID` quantiles across both countries.

Both approaches (section 4.3.3 and the shared-rank approach) achieve almost identical pooled cross-validated scores (0.2681 vs. 0.2731, see table 2), but the shared-rank approach reaches it with a single elastic-net per country rather than a four-model ensemble a parsimony argument and additionally provides an explicit cross-fold standard deviation (around 0.05), information the calibration-based approach did not provide. **This is the model shared rank across countries, elastic-net that is retained and submitted on the challenge platform**, and whose results are reported in section 5.

## 4.4 Validation protocol: cross-validation and a frozen holdout

Two levels of validation frame the approach above. The first is a 5-fold cross-validation, used throughout the iterations to compare approaches (sections 4.3.3 and 4.3.4). The second addresses a risk inherent to any iterative process: several successive rounds of changes were all validated on the same cross-validation split, and a trajectory in which every iteration improves the score on a fixed split looks as much like progressive overfitting to that split's noise as like genuine progress.

To settle this, roughly 20 % of the most recent days were set aside (a *lockbox*), never used in any cross-validation fold, any calibration, or any variable-selection decision. This score is computed only once and then cached: any subsequent change to the model reloads this frozen value instead of recomputing it, so that the out-of-sample check keeps its meaning and does not, through repeated re-reads, become a de facto third training signal. The results of this protocol are reported in section 5.2, alongside the score actually obtained on the platform.

# 5 Results

## 5.1 Performance of the reference model

Performance Détaillée (Cross-Validation 5 Folds)

| Pays | Moyenne_Spearman | Ecart_Type | Fold_1 | Fold_2 | Fold_3 | Fold_4 | Fold_5 |
|---|---|---|---|---|---|---|---|
| France | 0.1592 | 0.0546 | 0.0847 | 0.1710 | 0.2218 | 0.1250 | 0.1934 |
| Allemagne | 0.3575 | 0.0926 | 0.3831 | 0.3211 | 0.3709 | 0.2295 | 0.4829 |

Figure 7: Performance of the reference model (per-country XGBoost) in 5-fold cross-validation

These results obtained per country, before the move to the pooled framework of section 4.3 show a clear performance gap between France and Germany, consistent with H1. For Germany, the high mean correlation ($\sim$0.36) indicates that the model successfully orders price variations, suggesting that the non-linear relationships between prices, renewables, residual load, and fuels are well captured (H3, H4). Dispersion across folds remains moderate, despite higher variability tied to the intrinsically less stable nature of the German market. Conversely, the weaker performance in France ($\sim$0.16) reflects a harder-to-learn signal, consistent with the dampening role of nuclear generation. The reference model is thus more effective where the market is structurally more volatile and driven by variable physical constraints (Germany), while the stability of the French mix limits its predictive power in terms of relative price ranking. These two per-country scores do not, however, directly speak to the official pooled metric (section 4.3.2): that is the subject of the next section.

## 5.2 Performance of the retained model and the real score obtained

The final model shared rank across countries, one elastic-net per country, trained on the entirety of the training data (section 4.3.4) was used to generate a submission on the challenge platform. Table 2 brings together, on a comparable pooled scale, every estimate obtained along the trajectory of section 4.3, alongside the score actually obtained.

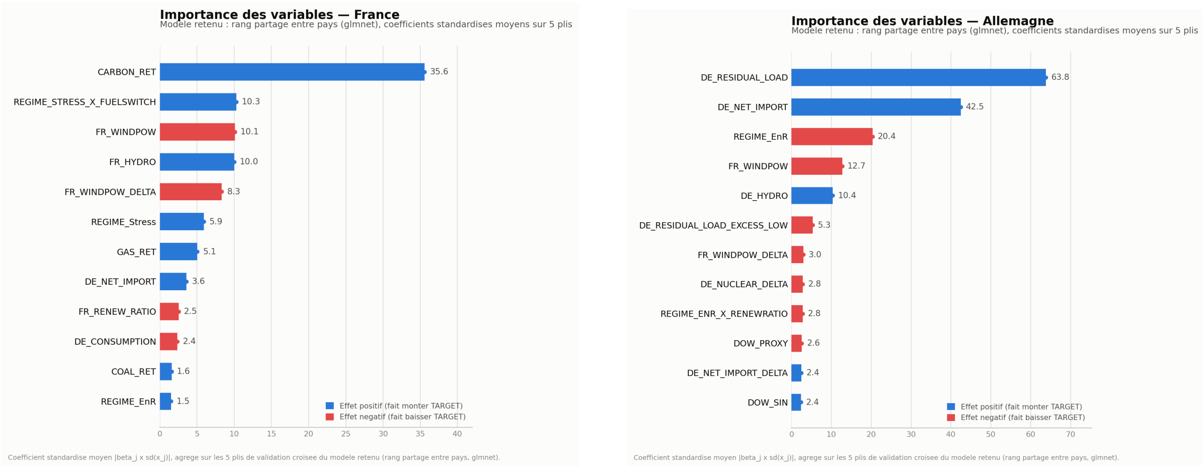
| **Measure** | **Spearman correlation** |
|---|---|
| Official QRT benchmark (public leaderboard) | 0.1586 |
| Four-model ensemble, calibrated, pooled (cross-validation) | 0.2681 |
| Shared rank across countries, pooled (cross-validation, 5 folds) | 0.2731 |
| Out-of-sample check (frozen lockbox, never revisited) | 0.2170 |
| **Real score obtained on the platform (final submission)** | **0.2268** |

Table 2: Comparison of internal estimates with the real score obtained on the challenge platform

The real score (0.2268) sits very close to the frozen lockbox (a gap of +0.0098) and clearly below the cross-validation estimate (a gap of $-0.0463$). This is a direct empirical validation of the protocol in section 4.4: cross-validation, consulted repeatedly over the course of the iterations, was indeed the optimistic measure, and the score one should honestly have expected was the more conservative of the two.

## 5.3 Variable importance of the retained model

The variable importance reported below describes the model actually submitted (shared rank across countries, section 4.3.4), not the reference model of section 4.3.1. It is computed directly on this model: at each fold of the 5-fold cross-validation used for the performance reported in table 2 (so never on the frozen lockbox, nor on the test data), the per-country elastic-net is refit and its standardized coefficient $\beta_j \times \mathrm{sd}(x_j)$ is extracted for each retained variable. Figure 8 aggregates the mean absolute value of this coefficient across the 5 folds a variable whose coefficient is zero in a given fold (the elastic-net's L1 penalty excluded it from that fold) contributes zero to the average for that fold, which naturally penalizes variables whose signal is not stable from one fold to another.



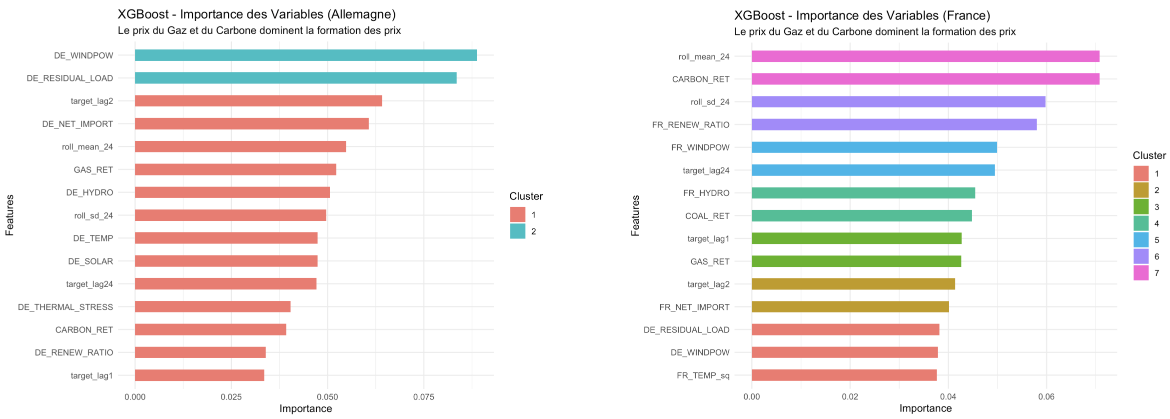

(a) France (b) Germany

Figure 8: Variable importance of the retained model (shared rank across countries, elastic-net), aggregated over the 5 cross-validation folds. Color encodes the sign of the effect (blue: pushes TARGET up; red: pushes it down), length encodes the magnitude of the mean standardized coefficient.

For France, `CARBON_RET` dominates by a wide margin (mean standardized coefficient of 35.6, more than three times the next variable), which directly confirms the economic intuition behind the new carbon variables tested in section 6 even though, as discussed in that same section, those new variables did not deliver any additional measurable gain beyond what `CARBON_RET` already captured. Next come the stress-regime × fuel-switch-spread interaction, French wind and hydro generation, consistent with H2 and H4: a reading driven by marginal production costs rather than demand volume alone. For Germany, `DE_RESIDUAL_LOAD` and `DE_NET_IMPORT` dominate by a wide margin (coefficients of 63.8 and 42.5), confirming H3/H4: the German market remains primarily driven by the physical supply-demand balance and cross-border flows with perfect sign consistency across the 5 folds for the dominant variables, which reinforces confidence in their role rather than in an artifact of a particular fold.

### 5.4 Comparison with the reference model

For comparison, figure 9 reports the variable importance of the reference XGBoost model (section 4.3.1) a result obtained at the initial submission, on a model that is no longer the one ultimately submitted, but which remains informative for understanding the signal exploited during the exploratory analysis.




(a) Germany (b) France

Figure 9: Variable importance of the reference model (per-country XGBoost)

In Germany, the system's physical variables wind, residual load, net flows already dominate in this reference model, as expected given the strong dependence on variable renewables and the market's sensitivity to balancing constraints (H3, H4). In France, market and temporal-inertia variables (rolling means and volatilities, carbon price) appear more important, reflecting a system more dampened by nuclear generation, where price dynamics are more smoothed over time. The ranking from the retained model (figure 8) differs from this one in the details expected, since the two models differ both in form (linear vs. trees) and in training objective (pooled rank vs. per-country rank) but converges on the essentials: a central role for `CARBON_RET` and temporal inertia in France, a central role for residual load and physical flows in Germany. This convergence, despite very different models, is an encouraging sign that these variables carry a genuine signal rather than an artifact of one particular method.

## 6 Verification of new variables: two hypotheses, one retained

Beyond the modeling trajectory of section 4.3, two additional feature-engineering ideas, motivated by H2 and H4, were tested after the initial submission. Each was first evaluated by its partial correlation with TARGET (controlling for what was already in the model), before any integration into the code to avoid adding a variable that looks relevant without actually contributing new information.

- **France rejected.** Interactions between CARBON_RET and the fuel-switch signal (FUEL_SWITCH_RET) were tested: near-zero partial correlation (0.031, not significant). A GAS_RET + CARBON_RET combination showed a higher marginal correlation, but this turned out to be an artifact: a linear combination of two variables already present adds nothing to a linear model, which can already reconstruct it on its own, and its strong correlation (0.851) with GAS_RET would in any case have exposed it to removal by the multicollinearity threshold of section 4.2.
- **Germany retained.** German residual load is negative nearly half the time in this sample (a surplus of wind and solar generation relative to load), precisely the regime where curtailment and the risk of negative prices appear (H3). A hinge-type variable (DE_RESIDUAL_LOAD_EXCESS_LOW), capturing the magnitude of this surplus, was tested and retained: partial correlation of $-0.161$ ($p < 0.0001$) after controlling for the residual load and the high-threshold indicator already present in the model a non-redundant signal.

Once integrated into the full pipeline, however, this last variable produced no measurable gain on the pooled score (a change of less than 0.01, below the inter-fold noise floor): an honest result to report as such rather than to oversell. This is a useful reminder that table 1 translates plausible economic hypotheses into things worth testing, not into guaranteed gains.

# 7 Discussion, limitations, and possible extensions

This work highlights structural mechanisms of price formation consistent with the five hypotheses of section 3: a French system dampened by nuclear generation (H1, H2) and a German market driven by physical constraints and variable renewables (H3, H4, H5). Moving from a per-country model to a pooled framework (section 4.3) made it possible to align with the metric actually used by the challenge, and the frozen out-of-sample validation protocol (section 4.4) made it possible to distinguish real progress from mere noise-fitting a distinction empirically validated by the gap between the cross-validation score and the real score obtained (section 5.2).
Several directions remain open to strengthen this approach. First, the persistent gap between the shared-rank approach (elastic-net, 0.2731 in cross-validation) and a full four-model ensemble on that same shared rank never tested for lack of a judged-sufficient gain at this stage remains a natural extension, especially since the real score

obtained suggests that cross-validation may also have underestimated the model's potential, not only its optimism. Second, more explicit integration of temporal dynamics, via multiple lag horizons, persistent regime indicators, or sequential models (LSTM, Temporal Convolutional Networks), would help better capture the inertia and anticipation effects specific to electricity markets particularly in France, where H1 suggests a slower-to-reveal signal. Third, the latent regimes identified in H3 could be directly integrated into the modeling, via regime-specific models rather than as a simple explanatory variable, to better capture the structural heterogeneity of market conditions. Finally, reformulating the objective price-quantile classification, pure ranking, or probabilistic prediction could be better suited to the leptokurtic, skewed nature of the target highlighted in H1.

# 8 Conclusion

This report starts from five hypotheses drawn from the exploratory analysis of French and German electricity prices (section 3) and follows their translation through to a validated, submitted model. The modeling trajectory (section 4.3) starts from a per-country reference model, identifies its structural limitation with respect to the challenge's pooled metric, and arrives at a shared-rank elastic-net across countries, simpler than a four-model ensemble for equivalent performance. Validated by a frozen out-of-sample protocol and submitted on the platform, this model achieves a real score of 0.2268, clearly above the official benchmark (0.1586). The resulting variable importance (section 5.3) confirms the initial economic hypotheses: a central role for the carbon price and temporal inertia in France, a central role for the physical supply-demand balance in Germany. Finally, the systematic verification of new variables before their integration (section 6) illustrates that an economically plausible hypothesis does not always translate into a measurable gain a distinction best established before writing the code rather than in a footnote afterward.